

Hello, Linux!

Before I start on the scientific tools, here are a few lines about setting up a Linux system for yourself, if you don't already have one. While bare-metal Linux installations are the best scenario, it may not always be possible, because your primary work may be on Windows. In such cases, the best solution is to go for a virtual Linux installation. In my humble opinion, I have found using VirtualBox to be a breeze when you need to set up a virtual machine. Just download a Linux distribution like Ubuntu and install it in a VirtualBox virtual machine. Once you have the basic installation done, take a little time to gain some hands-on familiarity with the terminal, explore the use of the package manager to install packages, and get comfortable using a text editor.

Here are some resources that can help you get started:

- Ubuntu community documentation on VirtualBox at <https://help.ubuntu.com/community/VirtualBox> (or <http://goo.gl/YGfR>)
- Documentation for the latest stable version of Ubuntu at <https://help.ubuntu.com/10.04/index.html> (or <http://goo.gl/BFMI>)

Giving LaTeX a try

LaTeX is a document preparation system for high quality typesetting, and is widely used for writing scientific papers, technical manuals, and even creating presentations (more on this later). TeX Live is an easy way to get up and running with LaTeX on your Linux system.

A file written in LaTeX is usually saved as a .tex file. You can edit LaTeX in your text editor of choice. The widely used VIM and Emacs both support LaTeX editing with syntax-highlighting, code-completion, etc. Once you have created your .tex file, you can compile the file using the *latex* command. This gives you a DVI file, which you can then convert to a PDF file using a tool like *dvipdf*. You can also directly get a PDF output using the tool *pdflatex*.

The best way to get started with LaTeX is to fire up your text editor, type in some LaTeX, and process it to see the results. Here are some great resources to help you on your way:

- The (not so) short introduction to LaTeX (<http://ctan.tug.org/tex-archive/info/short/english/short.pdf> or <http://goo.gl/wdIV>)
- Text processing using LaTeX (<http://www-h.eng.cam.ac.uk/help/tpl/textprocessing/> or <http://goo.gl/YLDX>)
- Draw the LaTeX symbol and get the corresponding command (<http://detexify.kirelabs.org/classify.html> or <http://goo.gl/maSx>)

Gnuplot

If you are doing experimental work, there is a fair chance that you are going to present your results in a graphical plot. Please welcome Gnuplot, a command-line-driven graphing utility for creating 2-dimensional as well as 3-dimensional graphical plots. It supports output to many file formats, with EPS and FIG being of special interest to the researcher. The easiest way to install Gnuplot is to use the package manager of your Linux distribution.

Here are a couple of resources to help you get started with Gnuplot:

- Gnuplot documentation (<http://www.gnuplot.info/documentation.html> or <http://goo.gl/IRqX>)
- Gnuplot's not so Frequently Asked Questions (<http://t16web.lanl.gov/Kawano/gnuplot/index-e.html> or <http://goo.gl/90xL>)

Calling Gnuplot from your C program

A cool feature of Gnuplot is that you can use it to display the continuously changing data that your program may be generating. For this example, let's assume that you have a program written in the C programming language, and you want to send the data it's generating to Gnuplot. The simplest way to do this is to use the *popen* system call. Please refer to the 2-cent tip published some time back at <http://linuxgazette.net/168/>


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